

main.py

Visualize

Run ▶

Output

```
1 def paths(m,n,N,i,j):
2     dp={(i,j):1}
3     ans=0
4     for _ in range(N):
5         nd={}
6         for (x,y),v in dp.items():
7             for a,b in [(x+1,y),(x-1,y),(x,y+1),(x,y-1)]:
8                 if 0<=a<m and 0<=b<n:
9                     nd[(a,b)]=nd.get((a,b),0)+v
10                else: ans+=v
11        dp=nd
12    return ans
13
14 print(paths(2,2,2,0,0))
15 print(paths(1,3,3,0,1))
```

6
12

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```
1 def climb(n):  
2     a, b = 1, 1  
3     for _ in range(n):  
4         a, b = b, a + b  
5     return a  
6  
7 print(climb(4))  
8 print(climb(3))
```

5
3

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```
1 def rob(a):  
2     def f(x):  
3         p = q = 0  
4         for n in x:  
5             p, q = q, max(q, p+n)  
6         return q  
7     return max(f(a[:-1]), f(a[1:]))  
8  
9     print(rob([2,3,2]))  
10    print(rob([1,2,3,1]))
```

3
4

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```
1 def paths(m, n):
2     dp = [1] * n
3     for i in range(1, m):
4         for j in range(1, n):
5             dp[j] += dp[j-1]
6     return dp[-1]
7
8 print(paths(7, 3))
9 print(paths(3, 2))
```



JS

TS